



DPUMOV

DPUMOV performs the simulation of pedestrians. That involves path planning, motion planning and collision avoidance of the simulated persons. Additionally, DPUMOV acts as a network server for the instances of DPUMOVClient.

DPUMOV can only run together with DPUSCNX. Additionally, either DPUTRFX or DPUTRFXClient should be available, so that the pedestrians can react to the EGO vehicle and the simulated traffic.

1. Inputs:

ExternalSimTime

External specification of the simulation time (used for of-line video generation)

In[1..10]

General purpose inputs for usage in MOVDetail control scripts. They are available as the variables i0 ... i9 in these scripts.

2. Dynamic parameters:

SlowSetupMS

Delay time per pedestrian [ms] for the first setup after entering an Area2 road segment.

Default value: 10, which means that 100 pedestrians per second are set up and the simulation is still running very smoothly.

Reasonable range: 0 to 100

TriggerTimeExponent

Parameter for weighting the previous trigger steps for the exponentially weighted moving average time measurement (*TriggerTimeEmwaMS*).

3. Static parameters:

NumPersons

Maximum number of simulated pedestrians in the "normal" category. Even if a module requests more than *NumPersons* pedestrians, they are not simulated. Depending on the CPU speed, the sum of *NumPersons* and *NumSpecialPersons* should not exceed 100 ... 200. Default: 50

NumSpecialPersons

Maximum number of simulated pedestrians in the "special" category. Even if a module requests more than *NumSpecialPersons* pedestrians, they are not simulated. Default: 50

PathPlanner

Default setting for the path planner algorithm, if it is not specified in the module.

(Possible values:

0 = Re-setup when reaching the target

1 = Just continue walking)

Default: 0



SILAB DRIVING SIMULATION

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<i>PreActivateDistance</i>	Distance (in meters), at which pedestrians are activated before entering the Area2 road segment in which they are defined. Default: 0 Reasonable range: 0 – 200 m.
<i>LeaveActiveNodes</i>	Number of Course elements, during which the pedestrians are left active, after the EGO vehicle has left their Area2 road segment. Default: 0 Reasonable range: 0 – 2
<i>SubSample</i>	Split pedestrian updates into several steps, so that each pedestrian is only planned in each <i>SubSample</i> 'th trigger step. The equation <i>SubSample</i> * <i>TriggerTime</i> should not evaluate to more than 100 ms. If no value for <i>SubSample</i> is specified, a reasonable value is automatically chosen.
<i>InformTRF</i>	Try to submit information on pedestrians to TRFX? Default value = <i>true</i> , but this will only work on the computer where DPUTRFX is running.
<i>ODB</i>	Boolean value: Insert pedestrians into the object database ODB? Default: <i>false</i>
<i>Threaded</i>	Boolean value: Perform path and motion planning in a separate thread? It is recommended to set <i>Threaded</i> = <i>true</i> if at least a quad-core processor is installed. Default: <i>false</i>
<i>Trace</i>	Configures Debug output. See section "Debug output".

4. Outputs:

<i>SimTime</i>	Current simulation time. It must be transmitted to the DPUMOVClient instances.
<i>Out[1..10]</i>	General purpose outputs for usage in MOVDetail control scripts. They are available as variables o0 ... o9 in these scripts.
<i>TriggerTimeMS</i>	Precise measurement of the time required for the last trigger step
<i>TriggerTimeEmwaMS</i>	Exponentially weighted average of the times required for the last trigger steps

5. Dependencies to other DPUs:

SILAB's pedestrian simulation consists of the DPUs DPUMOV, DPUMOVClient and DPUMOPSGE.

These DPUs depend on other DPUs as follows:

DPU	Variable
DPUMOV	DPUSCNX required DPUTRFX or DPUTRFXClient recommended
DPUMOVClient	DPUSCNX + Network connection to DPUMOV
DPUMOPSGE	DPUMOV / DPUMOVClient and DPUSGEWorld

For each dependency, please make sure that a higher Index is assigned to the dependent DPU than to its dependencies.



Usually, DPUMOV is started on the main simulation computer. The graphics computers run DPUMOVClient and DPUMOPSGE each.

For single-computer setups, only DPUMOV and DPUMOPSGE are required.

6. Functional principle:

For **single-computer setups**, the pedestrian simulation can be included as follows:

```
DPUMOV mov
{
    Computer = { WORKSTATION };
    Index = 56; # before MOPSGE

    NumPersons = 20;
};
```

```
DPUMOPSGE mopsg
{
    Computer = { WORKSTATION };
    Index = 57; # after MOV
};
```

For a **driving simulator** which consists of a main computer MAIN and three graphics computers VIS1 to VIS3, the pedestrian simulation can be integrated as follows:

```
DPUMOV mov
{
    Computer = {MAIN};
    NumPersons = 40;
    Index = 56;
};

DPUMOVClient movclient
{
    Computer = {VIS1, VIS2, VIS3};
    Index = 56;
    NumPersons = 40;
    ServerIP = "MAIN"; # Name of the computer on which DPUMOV runs
};

DPUMOPSGE mopsg
{
    Computer = { VIS1, VIS2, VIS3};
    Index = 57; # after MOV
};
```



```
Connections = {  
    mov.SimTime <-> movclient.SimTime  
};
```

7. Fixing problems:

If the pedestrians do not behave as expected, the reason is often that the pedestrian areas are not placed accurately enough or are blocked by static objects.

It is often helpful to use a schematic view of the simulation to spot these problems. SILABPreview's "birds-eye" view can be used for this purpose. Alternatively, DPUBirdsEye can be integrated into the simulation.

Please activate the parameters "DrawMovXXX" of the birds-eye perspective or of DPUBirdsEye to visualize the pedestrian simulation.

Detailed information about these options is available in DPUBirdsEye's documentation.

8. Debug output:

For easier debugging, the DPUMOV can output various debug messages. These are saved in the file SILAB.debug.log. The following message categories may be interesting for scenario developers:

<i>detevent</i>	Outputs all intermediate steps during the evaluation of control conditions and actions in MOVDetail scripts.
<i>manager</i>	Outputs messages when the simulation is started, paused or stopped
<i>animqueue pathqueue</i>	Outputs messages when errors in the animation of pedestrians occur (most of the cases, due to simulation lags)
<i>areadata</i>	Outputs messages when loading scenario data

Debug messages can be activated via the Trace parameter of DPUMOV. All message categories are concatenated and separated by spaces:

```
DPUMOV mov {  
    ...  
    Trace = "detevent manager areadata";  
    ...  
};
```

9. MOVDetail control:

Using a simple script language, the pedestrian's behavior can be influenced in more detail. For example, scenarios like "As soon as the EGO vehicle arrives at a certain point, a pedestrian emerges between two parked cars and crosses the street, so that the driver has to brake or evade in order to avoid an accident" can be realized. The MOVDetail control is described more precisely in the document "Reference Pedestrian simulation MOV".